Exercises: Convex Sets, Convex Functions, and Jensen's Inequality

Mathematical Foundations for Machine Learning

Convex Sets, Cones, and Convex Hulls

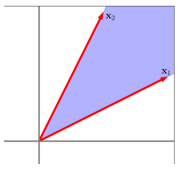
Exercise 1.1 [Easy]
Determine which of the following subsets of $\mathbb{R}^n$ are convex. Justify each answer.

(a) $S_1 = \{x \in \mathbb{R}^n \mid \|x\|_2 \leq 1\}$ (the closed unit ball).
(b) $S_2 = \{x \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$ (the unit circle).
(c) $S_3 = \{x \in \mathbb{R}^n \mid Ax \leq b\}$ for $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$ (a polyhedron).

Exercise 1.2 [Easy]
Let $K \subset \mathbb{R}^n$ be defined as $K = \{x \in \mathbb{R}^n \mid x_i \geq 0,\ i = 1, \ldots, n\}$ (the non-negative orthant).

(a) Show that K is a convex cone.
(b) Is K a proper cone? Justify your answer.

Pointed solid and closed
+ convex



sum inside
+scal inside

Exercise 1.3 [Medium]
Let $C_1, C_2 \subset \mathbb{R}^n$ be convex sets.

(a) Prove that $C_1 \cap C_2$ is convex.
(b) Give a counterexample showing that $C_1 \cup C_2$ need not be convex.
(c) The *Minkowski sum* is defined as $C_1 + C_2 = \{x + y \mid x \in C_1,\ y \in C_2\}$. Prove that $C_1 + C_2$ is convex.

Exercise 1.4 [Medium]
The convex hull of a set $S \subset \mathbb{R}^n$ is defined as

point in conv(S) = weighted combination of all points,
in boundary would be one single point with alfa = 1

$$\operatorname{conv}(S) = \Big\{ \sum_{i=1}^{k} \lambda_i x_i \;\Big|\; x_i \in S,\ \lambda_i \geq 0,\ \sum_{i=1}^{k} \lambda_i = 1,\ k \in \mathbb{N} \Big\}.$$

(a) Show that $\operatorname{conv}(S)$ is convex.
(b) Show that $\operatorname{conv}(S)$ is the smallest convex set containing S, i.e., if C is any convex set with $S \subseteq C$, then $\operatorname{conv}(S) \subseteq C$. z=(1k)w+kxk
(c) In the context of machine learning, suppose a neural network's weight space is $\mathbb{R}^d$ and we have a finite set of trained models $w_1, \ldots, w_k \in \mathbb{R}^d$. Describe geometrically what $\operatorname{conv}(\{w_1, \ldots, w_k\})$ represents. If the loss function $L(w)$ is convex, what can you say about the loss at any point in this convex hull?

$w = \sum_{i=1}^{k} \lambda_i w_i$

We take each weights as the average of all the models
that position conv({w1,…,wk}) -> if loss convex then bec.
of jensen the loss of the worst >= the hull's loss

Exercise 1.5 [Hard]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function. Define the *sub-level set*

$$L_\alpha(f) = \{x \in \mathbb{R}^n \mid f(x) \leq \alpha\}.$$

(a) Prove that $L_\alpha(f)$ is a convex set for every $\alpha \in \mathbb{R}$.
(b) Give a counterexample showing that the converse is false: a function can have all sub-level sets convex yet fail to be convex.

-e^x is non convex (hessian negative)

(c) Let $f_1, \ldots, f_m : \mathbb{R}^n \to \mathbb{R}$ be convex functions and define $g(x) = \max_{i=1,\ldots,m} f_i(x)$. Prove that g is convex. Then show that the feasible set

$$\mathcal{F} = \{x \in \mathbb{R}^n \mid f_i(x) \leq 0,\ i = 1, \ldots, m\}$$

is convex.

(d) In the context of training a neural network with m constraints on fairness metrics, each of the form $f_i(w) \leq 0$ where f_i is convex, explain why the feasible region of model parameters is convex even though the intersection of constraints may be complex.

Convex and Concave Functions — Characterizations

Exercise 2.1 [Easy]
Classify each of the following functions as convex, strictly convex, strongly convex, concave, or none. Justify each answer using the second-order condition where applicable.

(a) $f(x) = 3x^2 + 2x + 1$ on $\mathbb{R}$.
(b) $g(x) = -\log(x)$ on $\mathbb{R}_{++}$.
(c) $h(x_1, x_2) = x_1 x_2$ on $\mathbb{R}^2$.

Exercise 2.2 [Easy]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and $g : \mathbb{R}^n \to \mathbb{R}$ be concave.

(a) Show that $h(x) = f(x) - g(x)$ is convex.
(b) Is $-f$ concave? Prove or disprove.
(c) Let $\alpha \geq 0$. Show that αf is convex.

Exercise 2.3 [Medium]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable.

(a) State the first-order and second-order sufficient conditions for f to be convex.
(b) Using the second-order condition, prove that $f(x) = \|Ax - b\|_2^2$ is convex for any $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Under what condition on A is f strongly convex? Intereseting eigenvalues proof
(c) In Ridge regression, the objective is

$$L(\theta) = \|X\theta - y\|_2^2 + \lambda \|\theta\|_2^2$$

Adding L2 regularization makes it strongly convex no matter X

for $\lambda > 0$. Show that L is 2λ-strongly convex regardless of X.

Exercise 2.4 [Medium]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and differentiable.

(a) Prove the first-order characterization: f is convex if and only if

$$f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall\, x, y \in \mathbb{R}^n.$$

(Hint: For the forward direction, use the definition of convexity with parameter $t \in (0, 1]$ and $z = x + t(y - x)$, then take $t \to 0^+$.)

(b) Use (a) to show that if f is convex and differentiable, then any critical point (where $\nabla f(x^*) = 0$) is a global minimizer.

x - x0 grows as x0 - x does, so f(x) grows in both directions
-> by coercivity there is a minimum

Exercise 2.5 [Hard]
(a) Let $f : \mathbb{R}^n \to \mathbb{R}$ be m-strongly convex ($m > 0$). Prove that f has a unique global minimizer.

(Hint: Use the strong convexity inequality $f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{m}{2}\|y - x\|^2$ to show existence of a lower bound, and then prove uniqueness by contradiction.)

(b) Let f be strictly convex on a convex set $C \subseteq \mathbb{R}^n$. Show that f has at most one minimizer over C. Give an example showing that a minimizer need not exist (i.e., strict convexity does not guarantee existence without further assumptions).

(c) Consider the negative log-likelihood in logistic regression:

$$L(w) = \sum_{i=1}^{N} \Big[\log\big(1 + \exp(w^T x_i)\big) - y_i\, w^T x_i \Big],$$

where $y_i \in \{0, 1\}$ and $x_i \in \mathbb{R}^d$. Show that L is convex. Is it strongly convex in general? What common regularization technique makes the objective strongly convex, and why?

Exercise 2.6 **[Hard]**
Let $f_1, \ldots, f_m : \mathbb{R}^n \to \mathbb{R}$ be convex functions, and let $A \in \mathbb{R}^{n\times p}$, $b \in \mathbb{R}^n$.

(a) Prove that $g(x) = \sum_{i=1}^{m} \alpha_i f_i(x)$ is convex for any $\alpha_i \geq 0$ (non-negative weighted sum preserves convexity).

(b) Prove that $h(y) = f_1(Ay + b)$ is convex in y (composition with affine mapping preserves convexity).

(c) Prove that $\phi(x) = \max_{i=1,\ldots,m} f_i(x)$ is convex (pointwise maximum preserves convexity).

(d) Give a counterexample showing that the composition $f_1 \circ f_2$ of two convex functions is not necessarily convex. What additional condition on f_1 makes it work?

Jensen's Inequality and Applications

Exercise 3.1 **[Easy]**
Jensen's inequality states: if f is convex and X is a random variable with $\mathbb{E}[|X|] < \infty$, then

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

(a) Using $f(t) = t^2$, derive the inequality $\mathrm{Var}(X) \geq 0$ from Jensen's inequality.

(b) Using $f(t) = -\log(t)$ for $t > 0$, show that for positive values $a_1, \ldots, a_n$:

$$\frac{1}{n}\sum_{i=1}^{n} a_i \geq \left(\prod_{i=1}^{n} a_i\right)^{1/n}$$

(the AM-GM inequality).

Exercise 3.2 **[Medium]**
Let X be a random variable taking values in a convex set $C \subseteq \mathbb{R}^n$, and let $f : C \to \mathbb{R}$ be a **concave** function.

(a) State and prove Jensen's inequality for concave functions.

(b) Show that for any non-negative random variable X with finite mean:

$$\mathbb{E}[\sqrt{X}] \leq \sqrt{\mathbb{E}[X]}.$$

(c) In Bayesian machine learning, the *marginal log-likelihood* is $\log p(y) = \log \int p(y \mid w)p(w)\, dw$. Use Jensen's inequality applied to log (which is concave) to derive the *Evidence Lower Bound* (ELBO): for any distribution $q(w)$,

$$\log p(y) \geq \mathbb{E}_{q(w)}\left[\log \frac{p(y \mid w)p(w)}{q(w)}\right].$$

Exercise 3.3 [Hard]
(a) Let p and q be probability distributions over a finite set $\{1, \ldots, n\}$. The KL divergence is defined as

$$D_{\mathrm{KL}}(q\|p) = \sum_{i=1}^{n} q_i \log \frac{q_i}{p_i}.$$

Using Jensen's inequality applied to the convex function $f(t) = -\log(t)$, prove that $D_{\mathrm{KL}}(q\|p) \geq 0$ with equality if and only if $q = p$.

(b) Now consider the optimization problem

$$\min_q \; D_{\mathrm{KL}}(q\|p), \quad \text{s.t.} \quad q_i \geq 0, \; \sum_{i=1}^{n} q_i = 1.$$

Using part (a), what is the minimizer? Relate this to the entropy maximization problem

$$\max_q \; H(q) = -\sum_{i=1}^{n} q_i \log q_i, \quad \text{s.t.} \quad q_i \geq 0, \; \sum_{i=1}^{n} q_i = 1,$$

when p is the uniform distribution $p_i = 1/n$.

Epigraph, Subdifferential, Normal Cone, and Extreme Points

Exercise 4.1 [Medium]
(a) The *epigraph* of a function $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is defined as

$$\operatorname{epi} f = \{(x, t) \in \mathbb{R}^{n+1} \mid f(x) \leq t\}.$$

Prove that f is convex if and only if $\operatorname{epi} f$ is a convex set.

(b) Consider the *extended-value* function $f : \mathbb{R} \to \mathbb{R} \cup \{+\infty\}$ defined by $f(x) = -\sqrt{x}$ for $x \geq 0$ and $f(x) = +\infty$ for $x < 0$. Sketch $\operatorname{epi} f$ and determine whether f is convex.

Exercise 4.2 [Medium]
(a) Compute the subdifferential $\partial f(x)$ of $f(x) = |x|$ at $x = 0$, $x = 2$, and $x = -3$.

(b) Let $f(x) = \max(0, 1 - x)$ (the hinge loss used in SVMs). Compute $\partial f(x)$ for all $x \in \mathbb{R}$.

(c) State and verify the optimality condition $0 \in \partial f(x^*)$ for the function $f(x) = |x-3| + 2|x|$. Find all minimizers.

Exercise 4.3 [Medium]
(a) Let $C = \{x \in \mathbb{R}^n \mid \|x\|_2 \leq r\}$ be the closed ball of radius r. Compute the normal cone $N_C(x)$ for:

- $x \in \operatorname{int}(C)$ (i.e., $\|x\|_2 < r$),
- $x \in \partial C$ (i.e., $\|x\|_2 = r$).

(b) A point x in a convex set C is an *extreme point* if x cannot be written as $x = \lambda y + (1-\lambda) z$ for $y, z \in C$ with $y \neq z$ and $\lambda \in (0, 1)$. Find all extreme points of:

- The simplex $\Delta = \{x \in \mathbb{R}^n \mid x_i \geq 0, \; \sum x_i = 1\}$.
- The ℓ_∞-ball $B_\infty = \{x \in \mathbb{R}^2 \mid \|x\|_\infty \leq 1\}$.

Mixed Challenging Exercises

Exercise 5.1 [Hard]
Consider the *log-sum-exp* function $f : \mathbb{R}^n \to \mathbb{R}$ defined as

$$f(x) = \log\left(\sum_{i=1}^{n} e^{x_i}\right).$$

(a) Prove that f is convex using Jensen's inequality.
(Hint: Interpret f as $\log \mathbb{E}[e^X]$ for an appropriate discrete random variable.)
(b) Show that $\max_i x_i \leq f(x) \leq \max_i x_i + \log n$. (This shows that log-sum-exp is a smooth approximation to the max function.)
(c) Compute the gradient $\nabla f(x)$ and the Hessian $\nabla^2 f(x)$. Show that $\nabla^2 f(x) \succeq 0$, giving an alternative proof of convexity.
(Hint: The Hessian can be written as $\mathrm{diag}(s) - ss^T$ where $s = \mathrm{softmax}(x)$. Use Cauchy-Schwarz to show $v^T(\mathrm{diag}(s) - ss^T)v \geq 0$.)

Exercise 5.2 [Hard]
Consider the following optimization problem from statistical learning. Let $p \in \mathbb{R}^n$ with $p_i > 0$ and $\sum p_i = 1$ be a fixed probability distribution. Solve:

$$\min_{x \in \mathbb{R}^n} \sum_{i=1}^{n} x_i \left(\log x_i - \log p_i\right), \quad \text{s.t.} \quad x_i \geq 0, \ \sum_{i=1}^{n} x_i = 1.$$

(a) Show that the objective function $\phi(x) = \sum_i x_i(\log x_i - \log p_i)$ is strictly convex on the simplex $\{x \mid x_i > 0, \sum x_i = 1\}$.
(b) Write the Lagrangian with multiplier μ for the equality constraint and multipliers λ_i for the inequality constraints $-x_i \leq 0$.
(c) Apply the KKT conditions to find the optimal solution x^*.
(d) Interpret the result: what does it say about when the KL divergence $D_{\mathrm{KL}}(x\|p)$ is minimized?

Exercise 5.3 [Hard]
Let $A \in \mathbb{R}^{n \times n}$ be symmetric positive semidefinite and $b \in \mathbb{R}^n$.
(a) Consider the set

$$\mathcal{E} = \{x \in \mathbb{R}^n \mid x^T A x + b^T x \leq 1\}.$$

Prove that $\mathcal{E}$ is a convex set.
(b) Let $f : \mathbb{R}^n \to \mathbb{R}$ be a strongly convex function with parameter $m > 0$. Prove that for any minimizer x^* and any $x \in \mathbb{R}^n$:

$$f(x) - f(x^*) \geq \frac{m}{2}\|x - x^*\|^2.$$

(c) A data scientist is training a model by minimizing $L(w) + \frac{\lambda}{2}\|w\|^2$ where L is convex but not strongly convex, and $\lambda > 0$. The training algorithm reaches a point $\hat{w}$ with $f(\hat{w}) - f(w^*) \leq \epsilon$. Using (b), derive an upper bound on $\|\hat{w} - w^*\|$ in terms of ϵ and λ.

Exercises: Convex Optimization Problems

Mathematical Foundations for Machine Learning

Terminology and Conditions for Convex Optimization

Exercise 1.1 [Easy]
Consider the optimization problem

$$\min_{x\in\mathbb{R}^2} \; x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + x_2 \le 3, \quad x_1 \ge 0, \quad x_2 \ge 0.$$

(a) Identify the objective function, the inequality constraint functions, and the feasible set.

(b) Is this a convex optimization problem in standard form? Verify each of the three requirements (convex objective, convex inequality constraints, linear equality constraints).

(c) Is the feasible set convex? Justify your answer.

Exercise 1.2 [Easy]
For each of the following optimization problems, determine whether it is a convex optimization problem in standard form. If not, state which requirement is violated.

(a) $\min_{x\in\mathbb{R}} \; e^{-x} \quad \text{s.t.} \quad x^2 \le 1$

(b) $\min_{x\in\mathbb{R}^2} \; x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1^2 + x_2^2 = 1$

(c) $\min_{x\in\mathbb{R}^2} \; x_1 + x_2 \quad \text{s.t.} \quad x_1^2 + x_2^2 \le 4, \quad 2x_1 - x_2 = 1$

Exercise 1.3 [Medium]
Consider the problem

it is standard convex but not gradient in origin: 1D -> |x|, general grad: 0/0so ^2 is equ

$$\min_{x\in\mathbb{R}^n} \; \|x\|_2 \quad \text{s.t.} \quad Ax = b,$$

where $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$.

(a) This problem is not in standard convex form. Why not? (Hint: consider differentiability at the origin.)

(b) Show that this problem is equivalent to the convex optimization problem

$$\min_{x\in\mathbb{R}^n} \; \|x\|_2^2 \quad \text{s.t.} \quad Ax = b.$$

That is, both problems have the same set of optimal solutions.

(c) Verify that this reformulated problem is a convex optimization problem in standard form. Identify the problem class (LP, QP, QCQP, or SOCP).

Exercise 1.4 [Medium]
Consider the problem of minimizing the maximum component of a vector subject to linear constraints:

$$\min_{x\in\mathbb{R}^n} \; \max_{i=1,\dots,n} x_i \quad \text{s.t.} \quad Ax = b.$$

(a) Is this a convex optimization problem in standard form? Justify.

(b) Reformulate it as an equivalent **linear program** by introducing an auxiliary variable $t \in \mathbb{R}$. Write the LP in standard form.

(c) What general principle does this illustrate about equivalent convex problems?

min of piecewise-linearconvex function into LP:min

Local Optimality Implies Global Optimality

Exercise 2.1 [Easy]
Let $\Omega \subseteq \mathbb{R}^n$ be a convex set and $f : \Omega \to \mathbb{R}$ be a convex function. Suppose $x^* \in \Omega$ is a local minimizer, i.e., there exists $R > 0$ such that $f(x^*) \leq f(z)$ for all feasible z with $\|z - x^*\| \leq R$.

Prove that x^* is a global minimizer: $f(x^*) \leq f(y)$ for all $y \in \Omega$.

Exercise 2.2 [Medium]
global minimum: f(x) >= f(x*).

(a) Let $f : \mathbb{R}^n \to \mathbb{R}$ be strictly convex. Prove that if a global minimizer x^* exists, then it is unique.

(b) Give an example of a strictly convex function $f : \mathbb{R} \to \mathbb{R}$ that does *not* attain its global minimum. What additional condition guarantees existence?

(c) Show that if f is strongly convex with parameter $m > 0$, then f is coercive, i.e., $f(x) \to \infty$ as $\|x\| \to \infty$. Conclude that a strongly convex, lower semi-continuous function on a closed convex domain always has a unique global minimizer.

Exercise 2.3 [Hard]
convex -> all critical points global mins (one after the other)strictly -> at most one C

Let $f : \Omega \to \mathbb{R}$ be convex and continuously differentiable on a convex set Ω, and denote by $X \subseteq \Omega$ the feasible set.

A convex function implies continuity on interior points, not in boundary ->

we could have a convex funct. defined in two regions

(a) Prove that $x^* \in X$ is optimal if and only if

$$\langle \nabla f(x^*), y - x^* \rangle \geq 0, \quad \forall y \in X.$$

(Hint: use the first-order characterization of convexity for one direction, and a directional argument for the other.)

(b) Suppose now $X = \{x : Ax = b\}$. Show that the optimality condition reduces to: there exists $\nu \in \mathbb{R}^s$ such that $\nabla f(x^*) + A^T \nu = 0$.

(Hint: every feasible direction is of the form $y - x^* = v$ where $v \in \ker(A)$.)

Linear and Quadratic Programs

Exercise 3.1 [Easy]
Classify each of the following problems as an LP, QP, QCQP, SOCP, or none of these. Justify briefly.

(a) $\min_{x \in \mathbb{R}^3} \; 2x_1 - x_2 + 4x_3 \quad \text{s.t.} \quad x_1 + x_2 + x_3 = 1, \; x_i \geq 0$

(b) $\min_{x \in \mathbb{R}^2} \; x_1^2 + 3x_2^2 + 2x_1x_2 \quad \text{s.t.} \quad x_1 + x_2 \leq 5$

(c) $\min_{x \in \mathbb{R}^2} \; x_1 + x_2 \quad \text{s.t.} \quad x_1^2 + x_2^2 \leq 4, \; x_1^2 - x_2 \leq 0$

Exercise 3.2 [Medium]
A company produces n products. The profit per unit of product i is $c_i > 0$. Production of product i requires a_{ji} units of resource j, and there are b_j units of resource j available ($j = 1, \ldots, m$). Let x_i be the quantity produced of product i.

(a) Formulate the profit maximization problem as a linear program.

(b) Convert this to a minimization LP in standard form as defined in the lecture (with equality constraints and non-negativity).

(c) Now suppose the profit per unit is uncertain: c_i is a random variable with mean $\bar{c}_i$ and the cost vector c has covariance $\Sigma \succ 0$. The company wants to maximize the risk-adjusted profit $\mathbb{E}[c^T x] - \gamma \operatorname{Var}(c^T x)$ for some $\gamma > 0$. Formulate this as a QP.

Exercise 3.3 [Medium]
Consider Ridge Regression: given data $X \in \mathbb{R}^{N\times D}$, targets $y \in \mathbb{R}^N$, and regularization parameter $\lambda > 0$:

$$\min_{\theta\in\mathbb{R}^D} \|X\theta - y\|_2^2 + \lambda\|\theta\|_2^2.$$

(a) Show this is a QP by identifying the matrix P, the vector q, and the constant c in the standard form $\min \frac{1}{2}\theta^T P\theta + q^T\theta + c$.

(b) Prove that $P \succ 0$ for any $\lambda > 0$ and any X, and conclude that the problem is strictly convex with a unique minimizer.

(c) Derive the closed-form solution by setting the gradient to zero.

Exercise 3.4 [Hard]
Consider the Chebyshev center problem: find the center x_c and largest radius R of a Euclidean ball $B(x_c, R) = \{x_c + u : \|u\|_2 \le R\}$ contained in a polyhedron $P = \{x \in \mathbb{R}^n : \langle a_i, x\rangle \le b_i,\ i = 1, \dots, r\}$.

(a) Show that the constraint $B(x_c, R) \subseteq \{x : \langle a_i, x\rangle \le b_i\}$ can be written as

$$\langle a_i, x_c\rangle + R\|a_i\|_2 \le b_i.$$

(Hint: the ball lies in the half-space iff $\sup\{\langle a_i, x_c + u\rangle : \|u\| \le R\} \le b_i$.)

(b) Formulate the full optimization problem and show it is a linear program in the variables (x_c, R).

(c) Explain why this is a convex optimization problem even though we are *maximizing* R.

QCQP and SOCP

Exercise 4.1 [Medium]
Consider the robust linear program: given uncertainty sets $\mathcal{E}_i = \{\bar{a}_i + P_i u : \|u\|_2 \le 1\}$ with $P_i \in \mathbb{S}^n_+$, we want

$$\min_{x\in\mathbb{R}^n} c^T x \quad \text{s.t.} \quad \langle a_i, x\rangle \le b_i, \quad \forall\, a_i \in \mathcal{E}_i,\ i = 1, \dots, r.$$

(a) Show that the robust constraint $\langle a_i, x\rangle \le b_i$ for all $a_i \in \mathcal{E}_i$ is equivalent to

$$\langle \bar{a}_i, x\rangle + \|P_i x\|_2 \le b_i.$$

(b) Is this constraint linear, quadratic, or a second-order cone constraint? What problem class does the full robust LP belong to?

Exercise 4.2 [Hard]
Show the inclusion chain LP $\subset$ QP $\subset$ QCQP $\subset$ SOCP by answering:

(a) Explain why every LP is a QP. What is the matrix P in the quadratic objective?

(b) Explain why every QP is a QCQP.

(c) Show that a convex quadratic constraint $\frac{1}{2}x^T Px + q^T x + c \le 0$ with $P \in \mathbb{S}^n_+$ can be written as a second-order cone constraint. (Hint: use the Cholesky factorization $P = LL^T$ and complete the square.)

(d) For each inclusion, give a concrete example showing the inclusion is strict (i.e., a problem in the larger class that is *not* in the smaller class).

Mixed Challenging Exercises

Exercise 5.1 **[Hard]**
Consider the problem of finding the point in a convex polyhedron $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ closest to a given point $p \in \mathbb{R}^n$:

$$\min_{x \in \mathbb{R}^n} \|x - p\|_2^2 \quad \text{s.t.} \quad Ax \leq b.$$

(a) Classify this as an LP, QP, QCQP, or SOCP. Identify P, q, c in standard quadratic form.

(b) Prove that this problem always has a unique solution (state clearly which theorems you use).

(c) Use the optimality criterion for differentiable convex functions: show that x^* is optimal if and only if $\langle x^* - p,\, y - x^* \rangle \geq 0$ for all $y \in P$. Give a geometric interpretation of this condition.

(d) In the special case $P = \{x : x_1 + x_2 \geq 4,\ 2x_1 + x_2 \geq 5\}$ and $p = (0, 0)^T$, use the KKT conditions to find the closest point to the origin.

Exercise 5.2 **[Hard]**
In training a neural network, we minimize a loss $\ell(\theta)$ over parameters $\theta \in \mathbb{R}^D$ subject to a norm constraint to prevent overfitting:

$$\min_{\theta \in \mathbb{R}^D} \ell(\theta) \quad \text{s.t.} \quad \|\theta\|_2^2 \leq C.$$

(a) Suppose ℓ is convex. Is this a convex optimization problem? Which problem class (LP, QP, QCQP, SOCP, general convex) does it belong to? Justify.

(b) Suppose furthermore that ℓ is strictly convex. Using the results on convex optimization, what can you conclude about the minimizer?

(c) In practice, neural network losses are *non-convex*. Explain why the theorem "local min $\Rightarrow$ global min" does *not* apply, and give a simple one-dimensional example of a non-convex function with a local minimum that is not global.

(d) Despite non-convexity, practitioners use gradient descent and often find good solutions. Suggest one reason, from an optimization perspective, why this might work. (1–2 sentences suffice.)

Exercises: Duality Theory and KKT Conditions

Mathematical Foundations for Machine Learning

Lagrangian Formulation and KKT Conditions

Exercise 1.1 [Easy]
Consider the optimization problem

$$\min_{x \in \mathbb{R}^2} x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + x_2 \geq 2.$$

complementary slackness is g(x)=0so if g(x) activ

(a) Rewrite the constraint in the standard form $g(x) \leq 0$.
(b) Write down the Lagrangian $\mathcal{L}(x, \lambda)$.
(c) State all four KKT conditions for this problem explicitly (do not solve them yet).

Exercise 1.2 [Easy]
Consider the problem from Exercise 1.1. A candidate solution is $x^* = (1, 1)$ with multiplier $\lambda^* = 2$.

in convex local min -> global min

(a) Verify that all four KKT conditions are satisfied at (x^*, λ^*). if we find point that sat. all -> optim. point

(b) Is the objective function convex? Is the feasible set convex? What can you conclude about x^*?

(c) Give an interpretation of the value of $\lambda^* = 2$: if the constraint is changed to $x_1 + x_2 \geq 2 + \epsilon$ for small $\epsilon > 0$, by approximately how much does the optimal objective value change?

The bigger

Exercise 1.3 [Medium]
Formulate an optimization problem and use the KKT conditions to find the point in the set

$$M = \{x \in \mathbb{R}^2 : x_1^2 + x_2 \leq 4,\ x_1 - x_2 \leq 0\}$$

closest to the point $p = (0, 5)$, using the squared Euclidean distance as objective.

(a) Write the optimization problem in standard form.
(b) Write the Lagrangian and all KKT conditions.
(c) Use complementary slackness to determine which constraints are active, and solve for the optimal point.
(d) Verify that your solution is indeed a global minimizer.

Exercise 1.4 [Medium]
Consider the problem of minimizing the squared norm on the probability simplex:

$$\min_{x \in \mathbb{R}^n} \sum_{i=1}^{n} x_i^2 \quad \text{s.t.} \quad \sum_{i=1}^{n} x_i = 1, \quad x_i \geq 0\ \forall i.$$

(a) Write the Lagrangian with multiplier μ for the equality constraint and $\lambda_i \geq 0$ for the inequality constraints $-x_i \leq 0$.

(b) Write the KKT conditions and use complementary slackness to argue that $x_i^* = \frac{1}{n}$ for all i.

(c) What is the optimal value? How does it behave as $n \to \infty$? Give an interpretation in the context of diversification in portfolio optimization.

Exercise 1.5 [Hard]
Ridge regression as constrained optimization. Consider the problem

$$\min_{\theta \in \mathbb{R}^D} \|Y - X\theta\|_2^2 \quad \text{s.t.} \quad \|\theta\|_2^2 \leq t,$$

In the dual problem, instead of minimizing accounting for all the constraints, we are maximizing from below. We first compute all the possible minimizers for all constraints, and then get the constraint that gets the maximum minimizer. So we take that and we can say that it is smaller than the real minimizer of the feasible region (equal if strong duality).

If there is a feasible point that holds all the inequalities in a strict manner then the lagragian penalty mechanism can smoothly approximate it -> strong duality holds.

where $X \in \mathbb{R}^{N\times D}$, $Y \in \mathbb{R}^N$, and $t > 0$.

(a) Write the Lagrangian and derive the KKT conditions.

(b) From the stationarity condition, show that the optimal θ^* satisfies

$$\theta^* = (X^T X + \lambda I)^{-1} X^T Y$$

for some $\lambda \geq 0$, and explain why $X^T X + \lambda I$ is always invertible for $\lambda > 0$.

(c) Use complementary slackness to describe when $\lambda = 0$ (no regularization) vs. $\lambda > 0$ (active regularization). Relate this to the OLS estimator $\hat{\theta}_{\text{OLS}} = (X^T X)^{-1} X^T Y$.

(d) Prove that this constrained problem is convex and that strong duality holds (verify Slater's condition).

Dual Function and Dual Problem

Exercise 2.1 [Easy]
Consider the one-dimensional problem

$$\min_{x\in\mathbb{R}} x^2 \quad \text{s.t.} \quad x \geq 2.$$

(a) Rewrite in standard form and compute the Lagrangian.

(b) Derive the dual function $q(\lambda)$ by minimizing the Lagrangian over x.

(c) State the dual problem and solve it. What is d^*? What is p^*? Is there a duality gap?

Exercise 2.2 [Medium]
Dual of least-squares with linear constraints. Consider

$$\min_{x\in\mathbb{R}^n} \|x\|_2^2 \quad \text{s.t.} \quad Ax = b,$$

where $A \in \mathbb{R}^{p\times n}$ with $p < n$ and $b \in \mathbb{R}^p$.

(a) Write the Lagrangian $\mathcal{L}(x, \mu)$.

(b) Derive the dual function $q(\mu)$ by solving the stationarity condition $\nabla_x \mathcal{L} = 0$.

(c) Write the dual problem explicitly. Show it is a concave quadratic in μ.

(d) Assuming AA^T is invertible, solve the dual problem and recover the primal optimal solution x^*.

Exercise 2.3 [Medium]
Dual of a linear program. Consider the LP in inequality form:

$$\min_{x\in\mathbb{R}^n} c^T x \quad \text{s.t.} \quad Ax \leq b,$$

where $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$.

(a) Write the Lagrangian $\mathcal{L}(x, \lambda)$ with $\lambda \in \mathbb{R}^m$, $\lambda \geq 0$.

(b) Derive the dual function $q(\lambda) = \inf_x \mathcal{L}(x, \lambda)$. Identify the hidden constraint that must hold for $q(\lambda) > -\infty$.

(c) Write the dual problem explicitly, making the hidden constraint explicit. Show that the dual of a linear program is again a linear program.

Exercise 2.4 [Hard]
Dual of the Support Vector Machine. Consider the hard-margin SVM problem with training data $\{(x_i, y_i)\}_{i=1}^N$, where $x_i \in \mathbb{R}^D$ and $y_i \in \{-1, +1\}$:

$$\min_{w\in\mathbb{R}^D} \frac{1}{2}\|w\|_2^2 \quad \text{s.t.} \quad y_i(w^T x_i) \geq 1, \quad i = 1, \ldots, N.$$

(a) Rewrite the constraints in the form $g_i(w) \leq 0$ and write the Lagrangian.

(b) Derive the dual function by minimizing the Lagrangian over w. Express it purely in terms of the multipliers α_i and the data.

(c) Write the dual problem. Show that the dual depends on the data only through the inner products $\langle x_i, x_j \rangle$. Why is this important?

(d) State the complementary slackness conditions and explain why data points with $\alpha_i > 0$ are called *support vectors*.

(e) This problem is convex. Argue why Slater's condition holds if the data is linearly separable, and conclude that strong duality holds.

Weak Duality, Strong Duality, and Slater's Condition

Exercise 3.1 [Easy]
Consider the problem

$$\min_{x \in \mathbb{R}} (x-3)^2 \quad \text{s.t.} \quad x \geq 1.$$

(a) What is the primal optimal value p^*?

(b) Compute the dual function $q(\lambda)$ and evaluate it at $\lambda = 1$. Verify that $q(1) \leq p^*$ (weak duality).

(c) Prove the weak duality theorem in general: for any $\lambda \geq 0$, $q(\lambda) \leq p^*$.

Exercise 3.2 [Medium]
Consider the problem

$$\min_{x \in \mathbb{R}^2} e^{-x_1} \quad \text{s.t.} \quad \frac{x_1^2}{x_2} \leq 0, \quad x_2 > 0.$$

We restrict to the domain $\mathcal{D} = \{x \in \mathbb{R}^2 : x_2 > 0\}$.

(a) Show that the primal optimal value is $p^* = 1$.

(b) Compute the dual function $q(\lambda)$ and the dual optimal value d^*.

(c) Is there a duality gap? Does Slater's condition hold? Explain the connection.

Exercise 3.3 [Medium]
Consider the convex optimization problem

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad Ax = b, \quad g_i(x) \leq 0, \ i = 1, \ldots, r,$$

where f and all g_i are convex. Suppose Slater's condition holds (there exists $\tilde{x}$ with $A\tilde{x} = b$ and $g_i(\tilde{x}) < 0$ for all i).

(a) State the consequence of Slater's condition for this problem.

(b) Prove that for a convex problem with strong duality, if the primal optimum x^* and dual optimum (λ^*, μ^*) are attained, then the KKT conditions hold at (x^*, λ^*, μ^*).

(Hint: Use $f(x^*) = p^* = d^* = q(\lambda^*, \mu^*) = \inf_x \mathcal{L}(x, \lambda^*, \mu^*)$ and argue about the chain of inequalities.)

Exercise 3.4 [Hard]
When strong duality fails. Consider the non-convex problem

$$\min_{x \in \mathbb{R}} x \quad \text{s.t.} \quad x^2 \leq 0.$$

(a) Find the primal optimal value p^*.

(b) Compute the dual function $q(\lambda)$ and the dual optimal d^*.

(c) Show there is a duality gap and explain why. Which assumption of Slater's theorem is violated?

(d) Now consider the modified problem $\min_x x$ s.t. $x^2 - 1 \leq 0$. Show that Slater's condition holds, compute the dual function and dual optimal value, and verify strong duality.

Mixed Challenging Exercises

Exercise 4.1 [Hard]
Entropy maximization and KL divergence (Exam-style). Let $p = (p_1, \ldots, p_n)$ be a fixed probability distribution with $p_i > 0$ for all i. Consider the problem

$$\min_{x \in \mathbb{R}^n} \sum_{i=1}^{n} x_i\big(\log(x_i) - \log(p_i)\big) \quad \text{s.t.} \quad \sum_{i=1}^{n} x_i = 1, \quad x_i \geq 0 \; \forall i.$$

Note: this objective is the KL divergence $D_{\mathrm{KL}}(x \| p)$.

(a) Show that $\phi(x) = \sum_{i=1}^{n} x_i(\log x_i - \log p_i)$ is strictly convex on $\{x : x_i > 0\}$.

(b) Write the Lagrangian with multiplier μ for the equality constraint and $\lambda_i \geq 0$ for $-x_i \leq 0$. Derive all KKT conditions.

(c) Using complementary slackness and stationarity, show that $x_i^* = p_i$ for all i and determine the multipliers.

(d) What is the optimal value? Explain why this is consistent with the known property $D_{\mathrm{KL}}(q \| p) \geq 0$ with equality iff $q = p$.

(e) Assuming strong duality holds, derive the dual function and solve the dual problem. Verify that the dual optimal value equals the primal optimal value.

(f) *(Bonus)* The problem also has a saddle-point interpretation: $p^* = \inf_x \sup_{\lambda \geq 0, \mu} \mathcal{L}(x, \lambda, \mu)$. Without proof, state the corresponding expression for d^* and the condition under which they are equal.

Exercise 4.2 [Hard]
Constrained quadratic optimization with multiple constraints. Consider

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^T Q x + c^T x \quad \text{s.t.} \quad Ax \leq b, \quad Cx = d,$$

where $Q \in \mathbb{R}^{n \times n}$ is symmetric positive definite, $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $C \in \mathbb{R}^{p \times n}$, $d \in \mathbb{R}^p$.

(a) Write the Lagrangian $\mathcal{L}(x, \lambda, \mu)$ with $\lambda \in \mathbb{R}_+^m$ and $\mu \in \mathbb{R}^p$.

(b) Derive the dual function $q(\lambda, \mu)$ in closed form.

(c) Write the dual problem. Explain why it is always a concave maximization problem, regardless of the original problem's convexity.

(d) Prove that Slater's condition holds if there exists $\tilde{x}$ with $A\tilde{x} < b$ (componentwise) and $C\tilde{x} = d$. Conclude strong duality.

(e) For the specific instance $n = 2$, $Q = I$, $c = 0$, with constraints $x_1 + x_2 \leq 1$ and $x_1 - x_2 = 0$: use the KKT conditions to find the optimal solution, the active constraints, and all multipliers.

Exercise 4.3 [Hard]
Maximum likelihood with constraints. Consider N observations $y_1, \ldots, y_N$ drawn i.i.d. from a categorical distribution over $\{1, \ldots, K\}$ with probabilities $\theta = (\theta_1, \ldots, \theta_K)$. Let $n_k = |\{i : y_i = k\}|$ be the count of category k. The log-likelihood is

$$\ell(\theta) = \sum_{k=1}^{K} n_k \log \theta_k.$$

(a) Write the constrained MLE problem as a maximization with constraints $\sum_k \theta_k = 1$ and $\theta_k \geq 0$.

(b) Convert to a minimization problem. Write the Lagrangian and KKT conditions.

(c) Using complementary slackness, show that the MLE is $\hat{\theta}_k = n_k/N$.

(d) Derive the dual function and dual problem. Verify strong duality by showing the optimal dual value equals $-\ell(\hat{\theta})$.

(e) The dual of MLE connects to exponential families. Show that the stationarity condition can be written as $\mathbb{E}_{\hat{\theta}}[\mathbf{1}_{Y=k}] = n_k/N$, i.e., the model's expected sufficient statistics match the empirical ones. This is the *moment matching* property of MLEs in exponential families.